

Lecture 14

Interpolation of Spatial Data I

STAT 8020 Statistical Methods II

Background

Gaussian Process
Spatial Model

Spatial Interpolation

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Gaussian Process
Spatial Model

Spatial Interpolation

1 Background

2 Gaussian Process Spatial Model

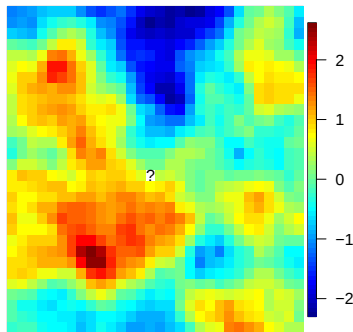
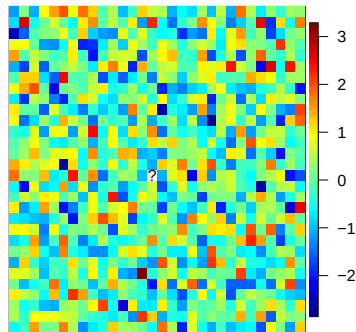
3 Spatial Interpolation

By the end of this module, you should be able to:

- Explain how spatial dependence can be exploited for interpolation and prediction
- Formulate Gaussian process models for spatially referenced data
- Understand the concepts of stationarity and isotropy and their implications for covariance modeling
- Identify commonly used covariance functions and characterize their smoothness and dependence properties
- Explain the principles of kriging as the best linear unbiased predictor
- Implement Gaussian process estimation, perform kriging prediction, and interpret associated uncertainty maps

Toy Examples of Spatial Interpolation

Consider two spatial images, each with a missing pixel denoted by “?”



Question: What is your best guess of the value of the missing pixel, denoted as $Y(s_0)$, for each case?

Discussion

Which image allows a more accurate prediction? Why? What information are you using?

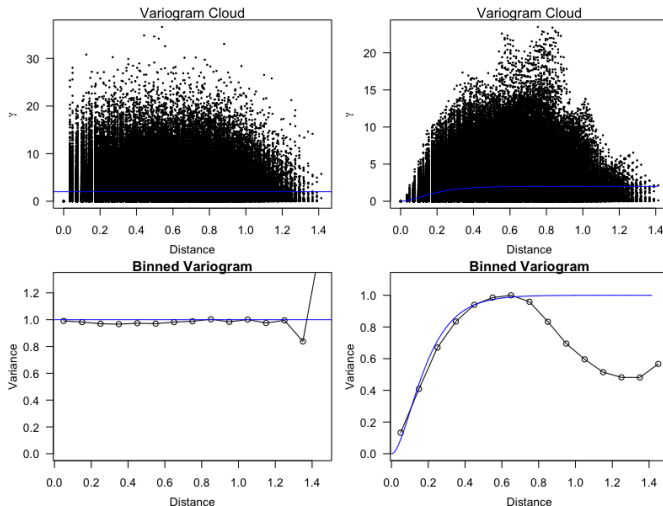
Visualizing Spatial Dependence Structure

Closer locations are more alike; the **variogram** measures how similarity decays with distance $\gamma(h) = \frac{1}{2} \text{Var}(Y(s) - Y(s+h))$

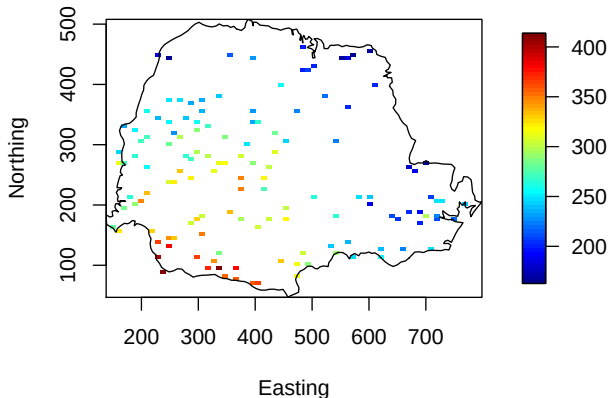
Background

Gaussian Process
Spatial Model

Spatial Interpolation



Interpolating Paraná State Precipitation Data



Challenge

Measurements are only available at monitoring stations.
How can we estimate precipitation everywhere else?

The Spatial Interpolation Problem

Given observations of a spatially varying quantity Y at n spatial locations

$$y(s_1), y(s_2), \dots, y(s_n), \quad s_i \in \mathcal{S}, i = 1, \dots, n$$

We want to estimate this quantity at any **unobserved location**

$$Y(s_0), \quad s_0 \in \mathcal{S}, \quad \hat{Y}(s_0) = E[Y(s_0) \mid \mathbf{Y} = \mathbf{y}]$$

⇒ The optimal predictor under **squared error loss** is the **conditional expectation**

Applications

- Mining: ore grade
- Climate: temperature, precipitation, ...
- Environmental Science: air pollution levels, ...

A Brief History of Spatial Statistics

● Applied Origins

- Mining [Krige, 1951], geology, and forestry: interpolation and resource estimation



● Geostatistics

- Variograms [Matheron, 1960s]. Matérn (1960): covariance models



● Statistical Foundations

- Cressie (1993)
- Stein (1999)



Spatial statistics evolved from geoscience applications to a rigorous statistical discipline based on stochastic processes, covariance modeling, and spatial prediction

Background

Gaussian Process
Spatial Model

Spatial Interpolation

1 Background

2 Gaussian Process Spatial Model

3 Spatial Interpolation

Linear Interpolation

The best guess (under squared loss) should be based on the conditional distribution $[Y(s_0) | Y = \mathbf{y}]$ where

$$\mathbf{y} = (y(s_1), \dots, y(s_n))^T$$

- Calculating this conditional distribution can be difficult
- Instead we use a **linear predictor**:

$$\hat{Y}(s_0) = \lambda_0 + \sum_{i=1}^n \lambda_i y(s_i)$$

- The best linear predictor is completely determined by the **mean** and **covariance** of $\{Y(s), s \in \mathcal{S}\}$

Why Linear?

Linear predictors are

- Computationally simple
- Optimal for Gaussian processes

Gaussian Process (GP) Spatial Model

Suppose the observed data $\{Y(\mathbf{s}_i)\}_{i=1}^n$ are measurements from an underlying spatial process

$$\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}},$$

where $\mathcal{S} \subset \mathbb{R}^d$ denotes the spatial domain.

Model: $Y(\mathbf{s}) = m(\mathbf{s}) + \epsilon(\mathbf{s}), \quad \mathbf{s} \in \mathcal{S}.$

- Mean function (trend):

$$m(\mathbf{s}) = \mathbb{E}[Y(\mathbf{s})] = \mathbf{x}^T(\mathbf{s})\boldsymbol{\beta},$$

which captures the large-scale spatial variation through spatial covariates

- Spatial residual process:

$$\{\epsilon(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}} \sim \text{GP}(0, K(\cdot, \cdot)),$$

where

$$K(\mathbf{s}_1, \mathbf{s}_2) = \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2))$$

is the covariance function describing spatial dependence

Covariance Modeling Assumptions: Stationarity and Isotropy

Residual process:

$$\epsilon(\mathbf{s}) \sim GP(0, K).$$

To estimate covariance K from the observed data, we typically impose stationarity and isotropy assumptions

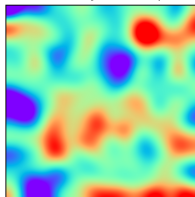
- Stationarity:

$$\begin{aligned} K(\mathbf{s}_1, \mathbf{s}_2) &= \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)) = C(\mathbf{s}_1 - \mathbf{s}_2) \\ &= \text{Cov}(\epsilon(\mathbf{s}_1 + \mathbf{h}), \epsilon(\mathbf{s}_2 + \mathbf{h})) \end{aligned}$$

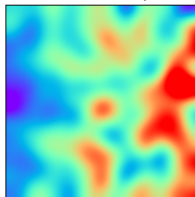
- Isotropy:

$$K(\mathbf{s}_1, \mathbf{s}_2) = \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)) = C(\|\mathbf{s}_1 - \mathbf{s}_2\|)$$

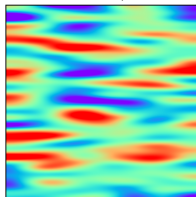
Stationary & Isotropic



Nonstationary



Anisotropic



A Valid Covariance Function Must Be Positive Definite!

A covariance function is positive definite (p.d.) if

$$\sum_{i,j=1}^n a_i a_j C(\mathbf{s}_i - \mathbf{s}_j) \geq 0$$

for any finite locations $\mathbf{s}_1, \dots, \mathbf{s}_n$, and for any constants a_i ,
 $i = 1, \dots, n$

Question: what is the consequence if a covariance function is NOT p.d.? $\Rightarrow$ **We can get a negative variance**

Question: How to guarantee a $C(\cdot)$ is p.d.?

- Using a **parametric covariance function** (see some examples in next slide)
- Using **Bochner's Theorem**  to construct a valid covariance function

Background

Gaussian Process
Spatial Model

Spatial Interpolation

Some Commonly Used Covariance Functions

Let $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ denotes the distance between two locations, σ^2 controls the marginal variance, and ρ controls the spatial range (how quickly dependence decays with distance)

- **Powered Exponential**

$$C(h) = \sigma^2 \exp\left\{-\left(\frac{h}{\rho}\right)^\alpha\right\}, \quad 0 < \alpha \leq 2.$$

- α controls smoothness
- $\alpha = 1$: Exponential covariance
- $\alpha = 2$: Gaussian covariance

- **Spherical**

$$C(h) = \sigma^2 \left(1 - \frac{3}{2} \frac{h}{\rho} + \frac{1}{2} \left(\frac{h}{\rho}\right)^3\right) 1_{\{h \leq \rho\}}.$$

- Compact support: covariance becomes exactly zero for $h > \rho$.
- Valid only for spatial dimensions $d = 1, 2, 3$

$$C(h) = \sigma^2 \frac{(\sqrt{2\nu}h/\rho)^\nu \mathcal{K}_\nu(\sqrt{2\nu}h/\rho)}{\Gamma(\nu)2^{\nu-1}}, \quad \sigma^2 > 0, \rho > 0, \nu > 0,$$

where

- ν : smoothness parameter
- $K_\nu(\cdot)$: modified Bessel function of the second kind

Why is the Matérn family popular?

- The smoothness parameter ν controls the differentiability of the spatial process.
- Exponential covariance when $\nu = 1/2$; Gaussian covariance as $\nu \rightarrow \infty$.
- The smoothness parameter ν allows more flexible modeling of spatial smoothness than the powered exponential model

“Use the Matérn model” – Stein (1999, pp. 14)

1-D Realizations from Matérn Model with Fixed σ^2, ρ

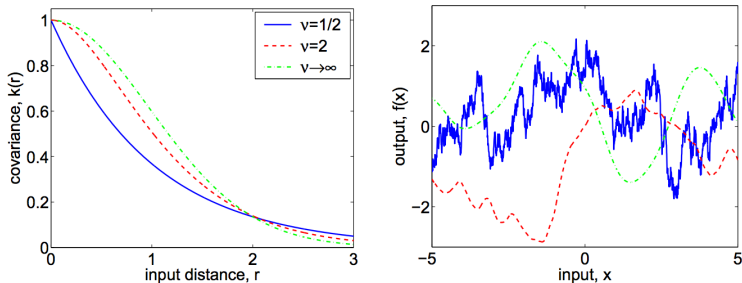


Figure: courtesy of Rasmussen & Williams 2006

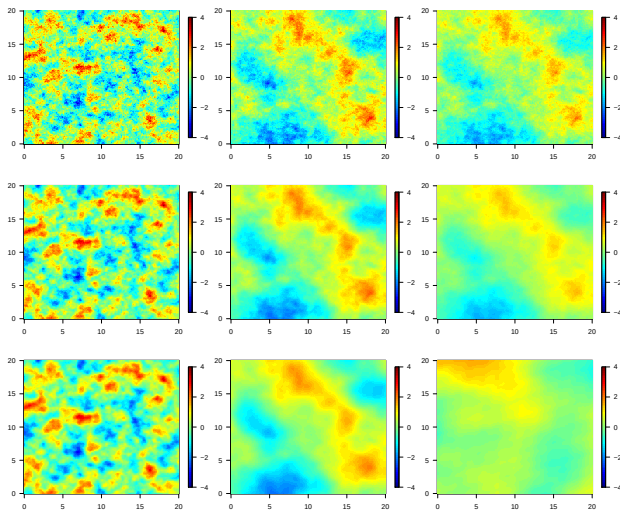
The larger ν is, the smoother the process is

2-D Realizations from Matérn Model with Fixed σ^2

Background

Gaussian Process
Spatial Model

Spatial Interpolation



Left $\rightarrow$ Right: increasing range (ρ)
smoothness (ν) σ^2 fixed

Top $\rightarrow$ Bottom: increasing

Background

Gaussian Process
Spatial Model

Spatial Interpolation

1 Background

2 Gaussian Process Spatial Model

3 Spatial Interpolation

If

$$\begin{pmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Then

$$[\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2] \sim N(\boldsymbol{\mu}_{1|2}, \Sigma_{1|2})$$

where

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{y}_2 - \boldsymbol{\mu}_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

Main Idea

Observed values provide information about unobserved values through the covariance structure

If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} m_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & k^T \\ k & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0 | \mathbf{Y} = \mathbf{y}] \sim N(m_{Y_0 | \mathbf{Y} = \mathbf{y}}, \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2)$$

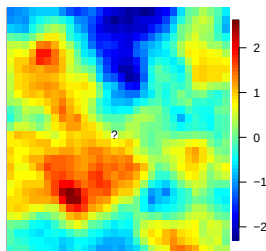
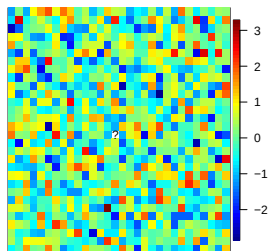
where

$$\begin{aligned} m_{Y_0 | \mathbf{Y} = \mathbf{y}} &= m_0 + k^T \Sigma^{-1} (\mathbf{y} - \mathbf{m}) \\ \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2 &= \sigma_0^2 - k^T \Sigma^{-1} k \end{aligned}$$

Next, we are going to revisit our toy examples

Toy Examples Revisited

For simplicity, we assume $m(s) = 0$ for $s \in \mathcal{S}$, the spatial covariance only depends on distance



$$m_{Y_0|Y=y} = 0 + k^T \Sigma^{-1} (y - 0), \quad \sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$$

Spatial uncorrelated field:

- $m_{Y_0|Y} = 0$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2$

Spatial correlated field:

- $m_{Y_0|Y} = k^T \Sigma^{-1} y$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$

Background

Gaussian Process
Spatial Model

Spatial Interpolation

Interpolating Multiple Points in Space

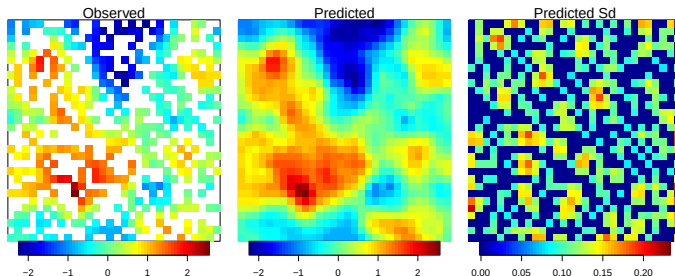
In practice, we would like to predict the values at many locations. The Gaussian conditional distribution formula can still be used:

$$[Y_0|Y = y] \sim N(m_{Y_0|Y=y}, \Sigma_{Y_0|Y=y})$$

where

$$m_{Y_0|Y=y} = m_0 + k^T \Sigma^{-1} (y - m)$$

$$\Sigma_{Y_0|Y=y} = \Sigma_0 - k^T \Sigma^{-1} k$$



If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} \mathbf{m}_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \Sigma_0 & \mathbf{k}^T \\ \mathbf{k} & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0 | \mathbf{Y} = \mathbf{y}] \sim N(\mathbf{m}_{Y_0 | \mathbf{Y} = \mathbf{y}}, \Sigma_{Y_0 | \mathbf{Y} = \mathbf{y}})$$

where

$$\mathbf{m}_{Y_0 | \mathbf{Y} = \mathbf{y}} = \mathbf{m}_0 + \mathbf{k}^T \Sigma^{-1} (\mathbf{y} - \mathbf{m})$$

$$\Sigma_{Y_0 | \mathbf{Y} = \mathbf{y}} = \Sigma_0 - \mathbf{k}^T \Sigma^{-1} \mathbf{k}$$

Question: what if we don't know $m(\mathbf{s}; \boldsymbol{\beta}), c(h; \boldsymbol{\theta})$?

⇒ The mean and covariance functions are typically unknown and must be estimated from the data $\mathbf{y}$. This will be covered in the next module

[Background](#)[Gaussian Process
Spatial Model](#)[Spatial Interpolation](#)

These slides cover:

- The problem of **spatial interpolation**
- **Stationarity** and **Isotropy** of a spatial process
- **Gaussian Process Spatial Models**

R functions/tricks to know:

- `vgram` (under the package `fields`) for visualizing spatial dependence
- `image.plot` (under the package `fields`) for visualizing spatial images
- Some matrix calculation tricks for speeding up computation

Bochner's Theorem

A complex-valued function C on $\mathbb{R}^d$ is the covariance function for a weakly stationary mean square continuous complex-valued random process on $\mathbb{R}^d$ if and only if it can be represented as

$$C(\mathbf{h}) = \int_{\mathbb{R}^d} \exp(i\omega^T \mathbf{h}) F(d\omega),$$

with F a positive finite measure. When F has a density with respect to Lebesgue measure, we have the spectral density f and

$$f(\omega) = \frac{1}{2\pi} \int_{\mathbb{R}^d} \exp(-i\omega^T \mathbf{h}) C(\mathbf{h}) d\mathbf{h}$$



Practical Takeaway

Bochner's theorem establishes a one-to-one relationship between stationary covariance functions and spectral densities. It provides a powerful tool for constructing valid covariance models and understanding spatial dependence across different spatial scales.